The Mortality Risk of Raising Grandchildren in the United States

by

Hongwei Xu
Department of Sociology, Queens College – CUNY

John R. Logan
Department of Sociology, Brown University

Todd K. Gardner
U.S. Census Bureau

CES 26-13 **February 2026**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

In the United States, grandparents who live with and provide primary care to their grandchildren have emerged as a particularly vulnerable group since the 1990s. Using confidential data from the U.S. Census Bureau and Social Security Administration, this study linked individuals aged 50 years or older from the 2000 census long-form sample to their death records from 2000–2019 (weighted n = 64,027,000) and examined the longitudinal association between coresident grandparenting status and mortality for non-Hispanic Whites, non-Hispanic Blacks, Hispanics, and Asians. We found consistently higher rates of mortality for White coresident grandparents and lower rates for Asian coresident grandparents, regardless of the duration of primary caregiving, compared to their peers without coresident grandchildren. We also found increased risks of mortality among Hispanic long-term primary caregivers but reduced risks among Black short-term primary caregivers, compared to their peers without coresident grandchildren.

Keyword: caregiver, coresident, grandparent, mortality, race, ethnicity

* Funding: This study was supported by a grant from the Professional Staff Congress at the City University of New York (Award # 67262-00 55) and a grant from the U.S. Social Security Administration (SSA) funded as part of the Retirement and Disability Research Consortium.

Disclaimer: The findings and conclusions expressed are solely those of the author(s) and do not represent the views of SSA, the U.S. Census Bureau, any agency of the federal government, or authors' affiliations. Neither the United States Government nor any agency thereof, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of the contents of this report. Reference herein to any specific commercial product, process or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply endorsement, recommendation or favoring by the United States Government or any agency thereof. The U.S. Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2939. (CBDRB-FY25-P2939-R11881; CBDRB-FY25-P2939-R11934; CBDRB-FY25-P2939-R11961; CBDRB-FY25-P2939-R12711).

INTRODUCTION

In the United States, it used to be unconventional for grandparents to live with grandchildren in the same households. Coresident grandparents did not attract much academic or public attention until the late 1990s, when the U.S. Census Bureau estimated that the number of children younger than 18 living in grandparent-maintained households increased from 2.3 million in 1980 to 3.9 million in 1997 (Bryson and Casper 1999) due to such factors as increased poverty, single parenthood, drug abuse, child abuse and neglect, and incarceration (Minkler 1999). In response to concerns about this trend, the Personal Responsibility and Work Opportunity Reconciliation Act of 1996 required the U.S. Census Bureau to collect data on caregiving grandparents (Simmons and Dye 2001). Later estimates showed that the number of grandparents living with grandchildren increased by 22% from 5.8 million in 2000 to 7.1 million in 2011–2013 (Florian and Casper 2015; Livingston 2013), then declined slightly to 6.7 million in 2021 (Anderson, Buck, and Hayward 2024).

Many coresident grandparents shoulder the burden of providing care to their grandchildren. An estimated 2.7 million coresident grandparents were primary caregivers to grandchildren in 2011 (Livingston 2013), although this number fell to 2.1 million in 2021. During the same period, however, the proportion of coresident grandparents who spent 5 years or more being the primary caregiver to their grandchildren increased from 39.3% (Ellis and Simmons 2014) to 49.3% (Anderson et al. 2024).

Living with and caring for grandchildren can affect grandparents' physical health, psychosocial health, and economic condition in various ways (Chen and Liu 2012; Luo et al. 2012; Minkler et al. 1997; Minkler and Fuller-Thomson 2001). In the U.S., studies reported negative health effects of caring for grandchildren. In particular, extensive and custodial

grandparenting has been associated with poor health outcomes including elevated depressive symptoms (Blustein, Chan, and Guanais 2004; Minkler et al. 1997), lower life satisfaction (Szinovacz, DeViney, and Atkinson 1999), and more functional limitations (Minkler et al. 2001). However, these health disadvantages may have preceded rather than resulted from grandparenting (Arpino and Bordone 2014; Di Gessa, Glaser, and Tinker 2016a; Di Gessa, Glaser, and Tinker 2016b; Hughes et al. 2007). Recent studies have suggested that the health implications of caring for grandchildren vary by grandparents' race and ethnicity (Chen et al. 2015; Choi 2020). For example, one regional study in Chicago reported that increased amount of time spent in grandchild care was associated with reduced feeling of loneliness and increased financial support from adult children among Chinese American grandparents (Kim 2022).

This study examined the risks of mortality among grandparents who not only live with but also are the primary caregivers of their grandchildren (Anderson et al. 2024; Bryson et al. 1999; Ellis et al. 2014; Florian et al. 2015; Fuller-Thomson, Minkler, and Driver 1997; Livingston 2013; Minkler and Fuller-Thomson 2000; Minkler and Fuller-Thomson 2005; Simmons et al. 2001). Drawing on nationally representative data from such sources as the decennial census, American Community Survey, and Current Population Survey, previous studies have well documented the patterns and trends of grandparents raising grandchildren. However, limited by the cross-sectional nature of these data, these studies could not assess any long-term health consequences for this vulnerable population. By matching participants in the 2000 U.S. census long-form sample to their death records between 2000 and 2019 from the Social Security Administration (SSA), this study was among the first to examine the association between coresident grandparenting and all-cause mortality.

A few studies have exploited longitudinal data, most notably from the Health and Retirement Study (HRS), to examine the mortality of grandparents who provide casual or intensive care to grandchildren (Choi 2020; Ellwardt, Hank, and Mendes de Leon 2021). However, their analytic samples were not large enough to study grandparents who are primary caregivers or racial and ethnic minorities such as Asian American grandparents. This study argues that the relationship between raising grandchildren and mortality is socially patterned by race and ethnicity. Capitalizing on the sheer size of the census long-form sample, this study examined racial and ethnic heterogeneity in mortality risk among caregiving grandparents across four groups: non-Hispanic Whites, non-Hispanic Blacks, Hispanics, and Asian Americans.

BACKGROUND

Grandchild Caregiving and Grandparents' Health

The extent to which grandparents are involved in providing care to grandchildren is affected by demographic, socioeconomic, and cultural factors, and these factors can directly affect the health and well-being of grandparents. Unlike occasional babysitting, providing primary care to grandchildren represents an extreme form of intergenerational support. In America, grandparents who are primary caregivers are more likely to be women and racial and ethnic minorities, have lower educational attainment, live in poverty, and have more functional limitations compared with grandparents who do not live with their grandchildren (Bryson et al. 1999; Ellis et al. 2014; Fuller-Thomson et al. 1997; Livingston 2013; Luo et al. 2012; Minkler et al. 2000; Minkler et al. 2005; Simmons et al. 2001). The socioeconomic disadvantages that predispose grandparents to providing primary care to grandchildren are stressors for caregivers and have deleterious effects on their health and well-being (Pearlin et al. 1990). Along with preexisting

socioeconomic disadvantages, providing care to grandchildren can directly affect grandparents' health and well-being through multiple pathways. Some pathways are positive, whereas others are negative.

On one hand, grandparents can experience role strains (Goode 1960) and chronic stress (Pearlin et al. 1990) when they struggle with caring for grandchildren due to limited family resources or inadequate coping strategies. The role strains of grandchild caregiving take many forms, ranging from physical and psychological to relational and financial, all of which have health consequences. Role strains can be particularly salient for grandparents raising grandchildren. As primary caregivers of grandchildren, they report frequent daily hassles, ranging from sleep deprivation to concerns about money for necessities and emergencies, and problems with both grandchildren and adult children (Mendoza, Fruhauf, and MacPhee 2020). Grandparents who are primary caregivers have much stronger feelings of stress, exhaustion, loneliness, and social isolation and report higher levels of depressive symptoms and psychological distress than their peers who have no or only partial grandchild caregiving responsibilities (Kelley 1993; Minkler et al. 2001). Financial burdens have been well documented in studies of grandparents who are primary caregivers. Many grandparents have already experienced lifelong economic hardship before they become primary grandchild caregivers. The sudden role transition to providing financial support to their young grandchildren can easily push these grandparents into poverty (Jendrek 1993; Minkler 1999; Waldrop and Weber 2001; Whitley and Fuller-Thomson 2017). The resulting economic strain increases the risks of depression and other forms of psychological distress among grandparents who are primary caregivers (Pearlin et al. 1990; Whitley et al. 2017).

On the other hand, according to role enhancement theory (Moen, Robison, and Dempster-McClain 1995), providing care to grandchildren allows grandparents to remain engaged in intellectual, social, and physical activities as they retire and withdraw from workplace social networks, experience an empty nest, or enter widowhood. By caring for and playing with grandchildren, grandparents can maintain a physically active lifestyle and adopt positive changes in health behaviors (e.g., participating in physical exercise while reducing smoking and drinking) in older age, which can boost their physical health (Di Gessa et al. 2016b; Hughes et al. 2007; Jendrek 1993; Sneed and Schulz 2019). More importantly, despite elevated risks of stress and psychological problems, providing care to grandchildren can be a source of positive emotions and an opportunity to establish rewarding family ties across multiple generations. Grandparents who are primary caregivers have reported multiple sources of life satisfaction and happiness in caregiving, including a new sense of purpose and direction for life, love and companionship of grandchildren, joy of helping grandchildren learn and grow, and a feeling of accomplishment of carrying on the family legacy (Dunifon, Musick, and Near 2020; Emick and Hayslip 1999; Waldrop et al. 2001). These positive emotions help grandparents mitigate the stress of primary caregiving and improve their mental health (Bachman and Chase-Lansdale 2005; Emick et al. 1999; Goodman et al. 2002).

Another potential health benefit of grandparenting pertains to cognition. Some caregiving activities are not physically demanding but rather cognitively stimulating. Examples include but are not limited to playing board games, reading books to young grandchildren, and helping school-age grandchildren with homework. Participation in these leisure and educational activities may help grandparents maintain cognitive functioning or slow cognitive decline in older age. Recent studies have consistently found that grandparents who provide care to grandchildren have

better cognitive performance in terms of episodic memory, verbal fluency, and executive function than their noncaregiving peers in the U.S. (Pan, Luo, and Bishop 2022; Sneed et al. 2019), Europe (Arpino et al. 2014), East Asia (Ahn and Choi 2019; Xu 2022), and Australia (Burn and Szoeke 2015).

Given both positive and negative pathways, the net effects of raising grandchildren likely depend on the context and circumstances (e.g., type, amount, intensity, and duration) of caregiving activities. Several studies have hypothesized a nonlinear relationship between the intensity of grandchild caregiving and grandparents' health (Arpino et al. 2014; Chen et al. 2015; Coall and Hertwig 2011; Di Gessa et al. 2016a; Di Gessa et al. 2016b; Hilbrand et al. 2017; Minkler et al. 2001; Xu et al. 2022). That is, the health benefits of grandparenting outweigh the strains when grandparents only provide casual or supplementary care to grandchildren. However, the burdens of grandchild caregiving escalate for grandparents as caregiving intensity increases, thereby reducing the health benefits. The health costs of grandchild caregiving gradually dominate the diminishing health benefits as caregiving intensity continues to rise and eventually exceeds a certain threshold. Therefore, using duration of primary caregiving as a proxy for the amount and intensity of caregiving activities, we explored the following hypotheses.

Hypothesis 1: Mortality risk is lower for grandparents who are short-term primary caregivers, compared with their peers without coresident grandchildren.

Hypothesis 2: Mortality risk is higher for grandparents who are long-term primary caregivers, compared with their peers without coresident grandchildren.

Racial and Ethnic Variations in Grandparenting

In the U.S., Black, Hispanic, and Asian American grandparents are more likely than their non-Hispanic White peers to provide casual, extensive, or even primary care to grandchildren. According to data from the U.S. Census Bureau, Black, Hispanic, and Asian grandparents have been consistently 2–4 times as likely as their White peers to provide primary care to grandchildren since 1990s (Anderson et al. 2024; Bryson et al. 1999). These patterns are related to not only racial and ethnic disparities in socioeconomic conditions but also differences in family norms, cultural expectations, and economic strategies with respect to grandparenting.

In general, non-Hispanic White grandparents are more likely to value their privacy, independence, and self-reliance and have a stronger preference for nuclear family households over multigenerational households, compared with racial and ethnic minorities (Cohen and Casper 2002; Kamo 2000). Therefore, they tend to live independently from adult children and grandchildren during the life course and avoid interference with parents' authority over their children (Casper and Bianchi 2002; Deaton and Stone 2014). When the middle generation experiences a life crisis, White grandparents may be compelled to invite adult children and grandchildren to live with them or even assume primary caregiving responsibilities for grandchildren. With limited coping skills and social support from relatives, friends, and community networks, they often find such living arrangements and caregiving roles extremely stressful. Nevertheless, compared with racial and ethnic minorities, White grandparents may have more economic resources to overcome the material hardship of raising grandchildren, thereby maintaining their own health and well-being.

In contrast, Black grandparents maintain a long-lasting tradition of surrogate parenting, rooted in West African culture and traditions of extended families. They are culturally accustomed to taking a stronger, more authoritative role in providing support, guidance, and

discipline to their grandchildren (Goodman et al. 2002; Jimenez 2002a; Minkler 1999). In particular, Black grandmothers have been hailed as kin-keepers and guardians of the generations since at least the enslavement period in the U.S. (Jimenez 2002b). Therefore, Black grandparents may be more adept at coping with the stress and strains of caregiving roles and receive stronger support from relatives, friends, and community networks, compared with their White peers. However, in contemporary American society, Black families have been disproportionately affected by, among other crises, the crack cocaine epidemic, AIDS epidemic, maternal incarceration, and teenage pregnancy. As a result, Black grandparents (especially grandmothers) are more likely than their White, Hispanic, and Asian peers to become primary caregivers of grandchildren as the unmarried or widowed heads of skipped-generation households, with both their spouse (if they ever married) and adult children absent (Minkler, Roe, and Price 1992; Minkler et al. 2005). These disadvantages in household structure exacerbate role strain and economic hardship for Black grandparents who are primary caregivers.

Unlike their White and Black peers who are mostly American-born citizens, Hispanic and Asian American grandparents are more likely to be immigrants. Thus, they share similar postimmigration challenges with respect to language barriers, adjustment to a new life in America with limited economic resources and access to public assistance, and the impacts of acculturation on traditional cultural values and practices (Burnette 1999; Minkler 1999). In the face of these challenges, Hispanic and Asian American grandparents prefer to live in multigenerational households maintained by adult children and care for grandchildren so that their children (especially daughters or daughters-in-law) can work full-time. In other words, Hispanic and Asian American grandparents may willingly choose to care for grandchildren in parent-headed multigenerational households as a resource-pooling strategy in response to

economic hardship. Such living arrangements also allow Hispanic and Asian American grandparents to adopt a coparenting approach and share grandchild caregiving burdens with adult children (Fuller-Thomson and Minkler 2007; Tang et al. 2019), as opposed to the authoritative role of surrogate parents played by Black grandparents in skipped-generation households.

Aside from economic necessity, both Hispanic and Asian American grandparents tend to embrace multigenerational coresidence and grandchild caregiving as normative practices that facilitate mutual aid, enhance intergenerational solidarity, and fulfill cultural obligations known as familism and filial piety (Casper et al. 2016; Glick and Van Hook 2002; Kamo 2000; Tang et al. 2019). Therefore, they may derive greater life satisfaction and improved mental health from grandchild caregiving, compared with their White and Black peers. In other words, the cultural factors of familism and filial piety can potentially mitigate the negative health effects of raising grandchildren and thus reduce the risk of mortality. It is worth noting that for Asian Americans, the traditional norm of filial piety also emphasizes old-age support provided by adult children to their aging parents. Thus, Asian American grandparents are more likely than their White, Black, and Hispanic peers to live with adult children and grandchildren as care recipients rather than caregivers (Kamo 2000; Tang et al. 2019).

These racial and ethnic differences in cultural preferences and practices of grandparenting may help explain the positive associations of casual or intensive grandchild caregiving with mental and physical health among racial and ethnic minority grandparents (Chen et al. 2015; Goodman et al. 2002). However, empirical evidence on racial and ethnic variations in grandparents as primary caregivers and their mortality is limited. Using data from the 1998–2014 waves of the HRS, Choi (2020) found that compared with noncaregivers, intensive (defined as more than 500 hours of babysitting in 2 years) noncoresident and multigenerational caregivers

were associated with lower mortality risks among White grandparents but higher mortality risks among Black grandparents. She found no association between any type of grandparenting (casual or intensive, coresident or noncoresident) and mortality among Hispanic grandparents.

Despite being oversampled, Black and Hispanic respondents, regardless of grandparent status, together account for less than a quarter of the HRS sample. The sample sizes of Black and Hispanic grandparents are even smaller, making it difficult to estimate the associations between grandparenting and mortality reliably. As for Asian Americans and other minorities, their sample sizes are too small to be analyzed (Choi 2020). Even for White grandparents in the HRS, sample size is substantially reduced when exploring heterogeneity in caregiving intensity and living arrangements. In Choi's (2020) analysis of the HRS data, for example, only 3.7% of White grandparents (n = 396) provided care to grandchildren in multigenerational households, and another 1.49% (n = 159) were caregivers in skipped-generational households, making regression estimates unreliable after adjusting for control variables.

Given this theoretical discussion, we expected racial and ethnic variations in the association between raising grandchildren and mortality.

Hypothesis 3: Providing primary care to grandchildren who live in the same household is more harmful to the longevity of White grandparents than to that of Black, Hispanic, or Asian American grandparents.

DATA AND METHODS

Data Sources

This study drew on two main data sources: the 2000 census long-form sample and the SSA's Numerical Identification file (Numident). The former features basic demographic and

socioeconomic information for about 17% of all American residents as of the 2000 decennial census. More importantly, as required by the Personal Responsibility and Work Opportunity Reconciliation Act of 1996, three questions were included in the 2000 census long form to collect information about coresident grandparents and their caregiving experience. Numident reflects all individual interactions with the SSA related to social security numbers (SSNs), including applications, claim records, requested changes to information, and death information, among others (Finlay and Genadek 2021). The U.S. Census Bureau obtains Numident data from SSA and processes quarterly updates from SSA transaction records to create a person-level data file to improve the quality of bureau surveys and decennial census data.

This study analyzed restricted person-level data from the 2000 census long-form sample and 2019 Numident file in Federal Statistical Restricted Data Centers. The restricted datasets are part of the Census Longitudinal Infrastructure Project, which assigns a unique, anonymous identifier, called the Protected Identification Key, to all individuals in the datasets available to the U.S. Census Bureau. Because the same person is assigned the same identifier across datasets, researchers can link individual records over time and across survey and administrative records. Results of analyses with these data are released only after a disclosure review by census professionals.

Analytic Sample

Consistent with previous research using survey data (Choi 2020; Ellwardt et al. 2021), the analytic sample in this study was restricted to participants in the 2000 census long-form sample who were aged 50 or older at the time of the census. Individuals who were younger than 50 might be too young to be grandparents or to die of natural causes. Using anonymous identifiers,

age-eligible individuals from the 2000 census long-form sample were linked to their records in the 2019 Census Numident file. This file includes every SSN holder from 1936, when SSNs were created and issued, to December 2019. Therefore, the analytic sample was further restricted to individuals who participated in the 2000 census long-form sample and had obtained their SSNs by December 31, 2019. Last, given the research focus in this study, the final analytic sample excluded individuals who were not non-Hispanic White, non-Hispanic Black, Hispanic, or Asian.

Dependent Variable

The outcome of interest in this study was all-cause mortality. The mortality information (including reported month, day, and year of death if deceased) included in Numident is collected by SSA to administer the Old-Age, Survivors, and Disability Insurance program, commonly known as Social Security. Mortality information in Numident is collected through multiple sources, including first-party reports of death from family members and representatives and verified third-party reports from friends, state government offices, Centers for Medicare and Medicaid Services, Department of Veterans Affairs, and Internal Revenue Service (Finlay et al. 2021). Using vital statistics from the Centers for Disease Control and Prevention (including the restricted-use National Death Index), the primary source of U.S. mortality data, as the benchmark, Finlay and Genadek (2021) concluded that Numident is a high-quality and timely (as fine as weekly) source of data to study all-cause mortality. This study assessed the risk of mortality as time to death, measured in days, from April 1, 2000, when participants were supposed to complete their census long forms. Time to death was right censored on December 31, 2019, if a person was still alive.

Independent Variables

All independent variables were measured in the 2000 census. The key independent variable in this study was coresident grandparenting status. In the 2000 census long-form sample, three questions were asked of all people aged 15 or older about their coresident grandparenting experience. The first question asked if the person was the grandparent of any grandchildren younger than 18 who were living in the same house. Those who answered "yes" were then asked if they were "currently responsible for most of the basic needs of one or more of these grandchildren." Grandparents who answered "yes" were further asked how long they had had this responsibility. The response categories included "less than 6 months," "6 to 11 months," "1 or 2 years," "3 or 4 years," and "5 years or more."

Combining responses to these three questions, each participant was classified into one of six groups. The first group included individuals who did not live with any grandchildren younger than 18 in the same house. These people could have young grandchildren living somewhere else, but the 2000 census long-form data were insufficient to identify their grandparent status. The second group consisted of grandparents who lived with their grandchildren but were not responsible for the basic needs of the coresident grandchildren. These grandparents could provide less intensive care to their coresident grandchildren, but the 2000 census long-form data did not capture any form of grandchild caregiving other than primary caregiving. The other four groups of participants were all referred to as grandparents raising grandchildren since they lived with and were primary caregivers to their grandchildren for varying durations. The original response categories of "less than 6 months" and "6 to 11 months" on the census long form were combined into a single category of "less than 1 year" to avoid small sample size, whereas the

other three categories remained unchanged. In a sensitivity analysis, the results for being a primary caregiver for less than 6 months were mostly not significant, suggesting the possibility of lack of statistical power due to small sample size.

Basic demographic characteristics included age (in years), gender (male versus female), and race and ethnicity. Combining their self-reported race and Hispanic origin, individuals were divided into one of four racial and ethnic groups: non-Hispanic White, non-Hispanic Black, Hispanic, and Asian. Household structure was captured by marital status (currently married or not), household head (or spouse), number of adults aged 18 or older, and number of children younger than age 18 living in the same household. Measures of socioeconomic status included educational attainment (less than high school, high school, some college, and bachelor's degree or above), employment (currently employed or not), income-to-poverty ratio, and housing status (homeowner versus renter). Nativity and acculturation were captured by citizenship status (born a U.S. citizen, naturalized citizen, and foreign national) and language (speaking English at home or not).

Last, grandparents who are unhealthy may be unwilling or less capable of caring for their grandchildren than their healthier peers. To account for potential self-selection by health, a dichotomous variable was created to indicate whether a person reported any of the following conditions: vision or hearing impairment, limitations in basic physical activities (e.g., walking, climbing stairs, reaching, lifting, or carrying), difficulty in self-care (e.g., dressing, bathing, or getting around at home), difficulty going outside the home alone (e.g., shopping or visiting a doctor's office), difficulty with mental capacity (e.g., learning, remembering, and concentrating), and difficulty working at a job or business.

Statistical Methods

Two sets of regression analyses were conducted. In the first set, individuals who were not coresident grandparents were used as the reference group to examine whether grandparents as primary caregivers had relatively higher risks of mortality. The second analyses were restricted to coresident grandparents only and used coresident grandparents who were not primary caregivers as the reference group. Two Cox proportional-hazard models were fitted in each set of regression analyses. Model 1 included grandparenting status as the key independent variable and only adjusted for basic demographic characteristics and state-fixed effects. Model 2 added other control variables to assess the extent to which the association between primary caregiving and mortality could be explained by household structure, socioeconomic status, acculturation, and health selection. To examine racial and ethnic variations in the association between raising grandchildren and mortality, all Cox models were fitted to each racial and ethnic group separately. Sampling weights were applied in model estimation to ensure national representativeness, and statistical significance of coefficients was evaluated based on robust standard errors clustered at the state level.

RESULTS

Descriptive Statistics

All descriptive statistics were weighted and all sample sizes reported in this study were rounded per disclosure requirements from the U.S. Census Bureau. Table 1 reports the frequence distributions of coresident grandparenting status in 2000, stratified by race and ethnicity. Substantial differences existed in the prevalence of coresident grandparents across racial and ethnic groups. Coresident grandparents were much more common among Hispanics (16.6%, $n \approx$

624,000), Asians (14.1%, $n \approx 252{,}000$), and Blacks (13.8%, $n \approx 754{,}000$) than among Whites (3.2%, $n \approx 1{,}702{,}000$). Among coresident grandparents, Blacks were most likely to be primary caregivers (49.2%), followed by Whites (39.1%) and Hispanics (31.6%); Asians were least likely to be primary caregivers (18.2%). Regardless of race and ethnicity, it was more common for grandparents to spend 5 years or more providing primary care to their grandchildren than any other shorter duration. Black grandparents spent more time providing primary care to their grandchildren than any other racial and ethnic group. For example, more than half (57.9%) of Black caregiving grandparents provided primary care to their grandchildren for 5 years or more, compared to only 36.9% of their Asian peers.

[Table 1 here]

Table 2 features descriptive statistics for the dependent variable, stratified by coresident grandparenting status, for Whites, Blacks, Hispanics, and Asians, respectively. Contrary to conventional wisdom, the crude death rate was not necessarily higher among coresident grandparents with or without primary caregiving responsibilities, compared with their peers without coresident grandchildren. In fact, among Whites, Blacks, and Hispanics, those who had no coresident grandchildren had higher crude death rates (56.0%, 57.5%, and 43.6%, respectively) than their counterparts who were primary caregivers of their grandchildren. Regardless of race and ethnicity, coresident grandparents without primary caregiving responsibilities also had higher death rates than grandparents raising grandchildren. Among grandparents raising grandchildren, long-term caregivers (5 years or more) had higher death rates (53.6%, 53.7%, 41.3%, and 40.8% for Whites, Blacks, Hispanics, and Asians, respectively) than short- or mid-term caregivers.

[Table 2 here]

Appendix Tables A1-A4 show descriptive statistics for the independent variables, stratified by coresident grandparenting status, for Whites, Blacks, Hispanics, and Asians, respectively. Starting with demographic characteristics, among Whites, Blacks, and Hispanics, the average age was similar between coresident grandparents without primary caregiving responsibilities and peers without coresident grandchildren. Among Asians, however, coresident grandparents without primary caregiving responsibilities were much older (mean age = 67.2 years) than grandparents raising grandchildren (mean age = 60.9–64.7) and peers without coresident grandchildren (mean age = 61.1). Regardless of coresident grandparenting status and race and ethnicity, more than half of the adults aged 50 or older were female.

In terms of household structure, coresident grandparents without primary caregiving responsibilities were the least likely to be married, the household head, or the spouse of the household head, and they also tended to live in larger households with more adults than grandparents raising grandchildren and peers without coresident grandchildren. Thess patterns were evident in every racial and ethnic group. Regardless of coresident grandparenting status, Blacks were much less likely to be married than any other racial and ethnic group.

Turing to socioeconomic status, older Americans without coresident grandchildren tended to have higher levels of education and income than coresident grandparents with or without primary caregiving responsibilities, regardless of race and ethnicity. Except for Asians, employment rate was lowest for coresident grandparents without primary caregiving responsibilities and highest for grandparents raising grandchildren. In each racial and ethnic group, there was no clear pattern of homeownership by coresident grandparenting status. Overall, regardless of coresident grandparenting status, Whites and Asians had relatively higher levels of education, income, and homeownership than Blacks and Hispanics.

In terms of nativity and acculturation, regardless of coresident grandparenting status, more than 90% of Whites and Blacks were born as U.S. citizens and spoke English at home. Depending on coresident grandparenting status, only about half or less of Hispanics and 10% or less of Asians were born as U.S. citizens, and more than 80% of these two racial and ethnic groups did not speak English at home. For both Hispanics and Asians, coresident grandparents were much more likely to be foreign nationals and less likely to speak English at home than their peers without coresident grandchildren. Thus, Hispanic and Asian coresident grandparents might maintain stronger cultural values from their countries of origin.

Last, there was limited evidence of health selection into living with or raising grandchildren, as measured by the rate of disability. Among Blacks and Hispanics, the rate of disability was quite similar between coresident grandparents without primary caregiving responsibilities and peers without coresident grandchildren, between coresident grandparents with and without primary caregiving responsibilities, and among grandparents raising grandchildren for different durations of time. For Whites and Asians, coresident grandparents had higher rates of disability than their peers without coresident grandchildren, and among grandparents raising grandchildren, those who spent 5 years or more providing primary care to their grandchildren had even higher rates of disability than those who spent less time on primary grandchild caregiving.

Regression Results

Table 3 reports the hazard ratios estimated from Cox models fitted to adults aged 50 or older for racial and ethnic groups separately. Only adjusting for age, gender, and state-fixed effects, Model 1 showed substantial racial and ethnic variations in the association between coresident

grandparenting status and mortality. Among Whites, regardless of the duration of primary care, grandparents raising grandchildren experienced more than 20% higher hazard rates of death than peers without coresident grandparents. Given that the analysis time is measured in days, the hazard rate can be interpreted as the probability of dying on any given day conditional on survival to the beginning of that day. About 50% of Whites aged 50 years or older died between 2000 and 2019 (see Table 2), so that the average number of deaths was approximately 3,800 per day and a 20% higher rate of death amounted to 760 deaths per day, which was nontrivial. The hazard rate of death tended to increase with the duration of primary caregiving, from 22.6% higher for less than a year of primary caregiving to 29.8% higher for 5 years or more. Coresident grandparents without primary caregiving responsibilities also had a 21.3% higher hazard rate of death than peers without coresident grandchildren.

[Table 3 here]

Compared with peers without coresident grandchildren, the hazard rate was 4% lower for Black grandparents who spent less than a year as primary caregivers and 4.7% higher for those who were primary caregivers for 5 years or more. Black grandparents who were primary caregivers for 1–2 years also had a lower rate of mortality, but this association was marginally significant. Black coresident grandparents without primary caregiving responsibilities did not have a significantly higher hazard of death than their peers without coresident grandchildren.

For Hispanics, there was a positive gradient in mortality by coresident grandparenting status. The hazard rate of mortality was 12.2% lower for Hispanic coresident grandparents without primary caregiving responsibilities and 6.2% lower for those who were primary caregivers for 1–2 years, compared with their peers without coresident grandchildren. However, being a primary caregivers for 5 years or more was associated with a 3.9% higher rate of death.

For Asians, living with or even raising grandchildren was generally associated with reduced risk of mortality. Compared with their peers without coresident grandchildren, Asian grandparents raising grandchildren experienced between 6.9% (primary caregiving for 5 years or more) and 13.3% (primary caregiving for 3–4 years) lower rates of death, depending on duration of caregiving. Asian coresident grandparents without primary caregiving responsibilities also experienced a 12.5% lower rate of death than their peers without coresident grandchildren. Only the coefficient for being a primary caregiver for less than a year was not statistically significant, but the point estimate suggested a lower rate of mortality.

After adding more control variables in Model 2, the associations between coresident grandparenting status and mortality remained substantively unchanged for Whites and Asians. For Whites, living with and raising grandchildren were still consistently related to higher rates of mortality. Compared with their peers without coresident grandchildren, the rate of mortality was 10% higher for coresident grandparents without primary caregiving responsibilities and even higher for grandparents raising grandchildren, ranging from 14.9% higher for less than a year of primary caregiving to 20.4% for 5 years or more. These results are consistent with Hypotheses 2 and 3.

For Asians, living with and raising grandchildren still had consistently lower rates of mortality in Model 2, supporting Hypotheses 1 and 3. The significance levels even increased for certain groups of grandparents as primary caregivers. For example, the association between being a primary caregiver for 5 years or more and mortality was marginally significant in Model 1 but highly significant ($p < .01$) in Model 2. This finding is contrary to Hypothesis 2 but consistent with Hypothesis 3.

For Blacks, raising grandchildren for less than a year or 1–2 years was still associated with a significantly lower rate of mortality in Model 2, lending support to Hypothesis 1. However, long-term (5 years or more) primary caregivers no longer experienced a significantly higher rate of death than their peers without coresident grandchildren. This loss of statistical significance is inconsistent with Hypothesis 2 but suggests that long-term Black grandparents who were primary caregivers might experience an elevated risk of mortality because of their disadvantages in household structure, socioeconomic status, acculturation, and disability status.

For Hispanics, providing short-term (1–2 years) primary care to grandchildren was no longer associated with lower rates of mortality in Model 2. By contrast, providing mid-term (3–4 years) and long-term (5 years or more) primary care to grandchildren were associated with 4.8% and 5.3% higher rates of mortality, respectively, compared with their peers without coresident grandchildren. These results are consistent with Hypothesis 2 but run counter to Hypothesis 3.

In terms of control variables, most coefficient estimates were consistent with the literature and in the same direction for all racial and ethnic groups. For example, men were more likely to die than women at any given age, whereas marriage and higher socioeconomic status were beneficial to longevity. It was noting that the hazard rate of mortality was lower for naturalized citizens and nationals but higher for those who spoke English at home, suggesting health selection among immigrants.

Next, we restricted the sample to coresident grandparents and compared risks of mortality across durations of primary caregiving. We used coresident grandparents without primary caregiving responsibilities as the reference group and reported the estimates of hazard ratios from weighted Cox models in Table 4. Overall, grandparents as primary caregivers did not experience any significantly higher risk of mortality than coresident grandparents without

primary caregiving responsibilities. This is not surprising because coresident grandparents can be diverse, ranging from being care recipients (from adult children or even grandchildren) to providing intensive care to grandchildren. White long-term (5 years or more) primary caregivers had a slightly higher rate of mortality, but this association was not statistically significant in Model 1 or Model 2. Hispanic long-term primary caregivers had a 12.6% higher rate of mortality in Model 1, but this association lost its significance after including additional control variables in Model 2. Black and Asian long-term primary caregivers had 1.6% and 3.8% higher rates of mortality, respectively, than coresident grandparents without primary caregiving responsibilities. According to the fully adjusted Model 2, providing short-term (less than a year or 1–2 years) primary care to grandchildren was associated with a significantly lower rate of mortality than simply living with grandchildren. This association was more evident among White and Black coresident grandparents than their Hispanic and Asian peers.

[Table 4 here]

Supplementary Analyses

Cox models assume that the hazard ratio is constant over time. We formally tested this assumption for every independent variable based on the scaled Schoenfeld residuals. It turned out that the assumption of proportional hazards was violated for most independent variables. This is not surprising given that the analysis time is measured in days over nearly 19 years (about 6,935 days in total). To assess the robustness of the Cox model estimates against the violation of the proportional-hazards assumption, we replicated the analyses reported in Tables 3 and 4 by fitting Gompertz models instead of Cox models. Appendix Tables A5 and A6 report the Gompertz modeling results for coresident grandparenting status. The estimated hazard ratios

from the Gompertz models are identical to those from the Cox models to at least the third decimal place. So are the statistical significance levels. The ancillary parameter gamma is positive and statistically significant in all the Gompertz models, but it was estimated to be no larger than 0.0003. In other words, the hazard rate indeed was not constant over time – it increased over time by no more than 0.03% a day, which roughly doubled over the entire period of 19 years.

To further explore the potential mechanism of cultural preferences of grandparenting, we stratified each racial and ethnic group by nativity (born as U.S. citizens or not). We reestimated Model 2 for each stratified subsample and reported the results in Appendix Tables A7 and A8. Among Whites, Blacks, and Hispanics, the significant associations between raising grandchildren and mortality found in Tables 3 and 4 were largely driven by those who were born U.S. citizens. The lack of significant associations for White, Black, and Hispanic foreign nationals may reflect diverse family values and cultural norms within these racial groups who are new immigrants (e.g., Whites from different European countries, Blacks from different African countries, and Hispanics from different Latin American countries). In contrast, the significant positive associations between raising grandchildren and mortality among Asians were largely driven by immigrants. Asian Americans are also diverse, but they are dominated by Chinese, Filipinos, Indians, Koreans, Vietnamese, and Japanese according to the census data. The significant findings among Asian immigrants and the lack of significant findings among Asian Americans who were born U.S. citizens together suggest that many Asian grandparents actively chose to immigrate to live with and care for their grandchildren (Tang et al. 2019).

DISCUSSION

American grandparents who live with and raise their grandchildren are generally associated with socioeconomic disadvantage because they face multiple challenges, including but not limited to caregiving burden, overcrowded housing, financial hardship, lack of social support, and aging-related health decline (Dunifon, Ziol-Guest, and Kopko 2014). Given their socioeconomic disadvantages, it is not surprising that grandparents who are primary caregivers often struggle with both physical and mental health problems. However, grandparents raising grandchildren are known to be diverse in terms of race and ethnicity, culture, and family circumstances that give rise to intergenerational coresidence and caregiving (Hayslip and Kaminski 2005; Hayslip, Fruhauf, and Dolbin-MacNab 2019). By linking participants in the 2000 census long-form sample to their death records for the following 2 decades, we found that consistent with Hypothesis 3, the relationship between raising grandchildren and mortality was socially patterned by race and ethnicity, after adjusting for demographic, socioeconomic, acculturation, and health factors.

White grandparents who were primary caregivers experienced increased risk of mortality, compared with their peers without coresident grandchildren. The risk of mortality also tended to rise as the duration of primary caregiving increased, supporting Hypothesis 2. These findings are not surprising given multigenerational coresidence and extensive grandchild caregiving are often the result of family crises and negatively appraised, if not stigmatized, among White grandparents (Pruchno 1999; Pruchno and McKenney 2002). Despite their socioeconomic advantages, White grandparents had difficulty overcoming the role strain of living with and raising grandchildren. These findings are also in line with previous research reporting worse physical and mental health outcomes among White coresident grandparents (Chen et al. 2015; Goodman and Silverstein 2006; Pruchno et al. 2002; Szinovacz et al. 1999). However, our

findings are different from Choi's (2020) analysis of HRS data, which reported a lower risk of mortality for White coresident caregiving grandparents. This discrepancy may be attributed to different definitions of caregiving grandparents. Choi defined grandparent caregivers as those who spent at least 100 hours in the past 2 years taking care of grandchildren. This definition is different from the grandparents who are responsible for the basic needs of their coresident grandchildren in our study.

For Blacks, we initially found a higher risk of mortality among long-term (5 years or more) primary caregivers, compared with their peers without coresident grandchildren, but this association lost its statistical significance after adjusting for control variables other than basic demographic characteristics. In fact, we found a significantly lower risk of mortality among short-term (2 years or less) Black primary caregivers, compared with their peers without coresident grandchildren. These findings are consistent with the observation that Black grandparents raising grandchildren often face socioeconomic disadvantages but are culturally accustomed to playing the roles of kin-keeper and guardian of the generations (Jimenez 2002b). As a result, they are better at seeking social support and coping with the psychological challenge that arises from primary caregiving. Compared with their White counterparts, Black grandparents may have worse physical health (Chen et al. 2015), but they enjoy greater life satisfaction and less psychological distress and depressive symptoms (Bachman et al. 2005; Goodman et al. 2002; Goodman et al. 2006; Pruchno et al. 2002). In other words, Black grandparents' mortality disadvantage may be largely explained by their preexisting disadvantage in socioeconomic status and associated poor physical health (Chen et al. 2015; Goodman et al. 2002; Goodman et al. 2006).

For both Hispanic and Asian grandparents, living with and providing care to grandchildren in multigenerational households represent more than an economic strategy to combat poverty. These living arrangements and caregiving practices are culturally preferred because they strengthen mutual aid and intergenerational solidarity in the extended family (Dunifon et al. 2014; Fuller-Thomson et al. 2007; Kamo 2000; Tang et al. 2019). However, we found different mortality consequences of raising grandchildren between Hispanic and Asian grandparents. We initially found lower risks of mortality among Hispanic coresident grandparents without primary caregiving responsibilities and short-term primary caregivers, compared with their peers without grandchildren, but these advantages were fully attenuated after controlling for household structure, socioeconomic status, acculturation, and disability. We also found significantly higher risks of mortality among mid- and long-term Hispanic primary caregivers, and these disadvantages persisted in the fully adjusted model. These findings together suggest that despite some psychological benefits reported in previous studies of Hispanic grandparents (Goodman et al. 2002; Goodman et al. 2006), the role strain of raising grandchildren for a prolonged period could still take a toll on their longevity.

In contrast, we found overwhelmingly lower risks of mortality among Asian grandparents living with and raising grandchildren, compared with their peers without coresident grandchildren. In our analytic sample, despite being older and predominantly immigrants, Asian grandparents living with and raising grandchildren had relatively higher socioeconomic status, measured by educational attainment, income, and homeownership, than Blacks and Hispanics. Nevertheless, the mortality advantage persisted for these Asian grandparents after controlling for their socioeconomic status, household structure, acculturation, and disability status. Several studies based on qualitative interviews or small convenience samples have reported that Asian

grandparents view their role of grandchild caregivers more as a normative responsibility that enhances mutual reciprocity, family unity, and closeness rather than a burden (Kataoka-Yahiro, Ceria, and Yoder 2004; Kataoka-Yahiro 2010). As a result, Asian grandparents tend to report greater role satisfaction and better self-rated health when they are involved in grandchild caregiving. Even when they provide extensive care (more than 30 hours per week) to grandchildren, Asian grandparents do not report their overall health status is compromised by the burden of caregiving (Kataoka-Yahiro et al. 2004; Kataoka-Yahiro 2010). Perhaps the positive cultural values associated with grandparenting and support from their extended family empower Asian grandparents to be resilient and resourceful to the extent that they overcome the role strain of raising grandchildren.

These findings should be interpreted in light of several study limitations. First, there may have been negative health selection into living with and caring for grandchildren among grandparents. The 2000 census long-form data only allowed us to adjust for physical and mental disabilities. Our descriptive statistics (Appendix Tables A1-A4) show that grandparents raising grandchildren had similar or even higher rates of disability than their peers without coresident grandchildren. Nevertheless, we cannot rule out selection bias due to other forms of health conditions such as chronic diseases and mild cognitive impairment. In presence of positive health selection, we would have overestimated the positive association between raising grandchildren and mortality among Asian Americans and underestimated the negative association among White Americans.

Second, all independent variables were measured at baseline even though many of them were time-varying variables. For example, it is possible that a primary caregiver at baseline withdrew from caregiving in later years or an older adult who was not a grandparent at baseline

became a caregiving grandparent in later years. We could not measure such time-varying dynamics, although our supplementary analyses indicate time-varying hazards for coresident grandparents with or without primary caregiving responsibilities. If the grandparents raising grandchildren in 2000 eventually withdrew from their caregiving responsibilities as their grandchildren grew older, and other older adults became responsible for grandchildren after 2000, our estimates of the association between raising grandchildren measured in 2000 and mortality over the period of 2000-2019 would be biased towards null. Hence, we may have underestimate the mortality consequence of raising grandchildren.

Third, the two reference groups used in Tables 3 and 4, respectively, are heterogeneous. Those who had no coresident grandchildren could have grandchildren living somewhere else or could have no grandchildren at all, while coresident grandparents without primary caregiving responsibilities could be heavily involved in grandchild care or could be care recipients themselves. The census data are unable to capture such heterogeneous grandparenting experiences, making it difficult to interpret our results about primary caregiving. Nevertheless, we could clearly identify and gain new insights into grandparents as primary caregivers by comparing them against coresident grandparents without primary caregiving responsibilities.

Last, the Numident data only include individuals who have applied for SSNs. Therefore, our study excluded certain marginalized populations such as undocumented immigrants. Furthermore, the Numident data record the date of death but not cause of death, limiting our ability to explore potential mechanisms underlying the association between coresident grandparenting and mortality. Last, we had to group all Hispanic subgroups and all Asian subgroups due to small subsample sizes for coresident grandparents. Future research is needed to

explore within-group heterogeneity among Hispanic grandparents (Fuller-Thomson et al. 2007) and Asian grandparents (Tang et al. 2019).

Despite these limitations, this study shed new light on racial and ethnic diversity in the experience of coresident grandparenting and associated mortality consequences. It refutes the stereotype of coresident grandparents as a homogeneous and vulnerable population by highlighting the mortality advantage of racial and ethnic minority grandparents living with and raising grandchildren. Future research is needed to explore the strengths of grandparents raising grandchildren in the face of adversity, and public policymakers should consider racial and ethnic heterogeneity in supporting these grandparents. For White grandparents, an educational campaign that emphasizes the diversity and benefits of grandchild care may help them cope with social stigma and find mutual support. Long-term primary caregivers, especially among Blacks and Hispanics, should receive priorities in child welfare programs such as the Child Tax Credit and public preschools to reduce their caregiving burden.

References

Ahn, Taehyun and Kyong Duk Choi. 2019. “Grandparent Caregiving and Cognitive Functioning among Older People: Evidence from Korea.” *Review of Economics of the Household* 17(2):553–86.

Anderson, Lydia R., Clayton Buck, and George M. Hayward. 2024. “Grandparents and Their Coresident Grandchildren: 2021.” *Current Population Reports*: P20-588. U.S. Census Bureau.

Arpino, Bruno and Valeria Bordone. 2014. “Does Grandparenting Pay Off? The Effect of Child Care on Grandparents' Cognitive Functioning.” *Journal of Marriage and Family* 76(2):337–51.

Bachman, Heather J. and P. Lindsay Chase-Lansdale. 2005. “Custodial Grandmothers’ Physical, Mental, and Economic Well-Being: Comparisons of Primary Caregivers from Low-Income Neighborhoods.” *Family Relations* 54(4):475–87.

Blustein, Jan, Sewin Chan, and Frederico C. Guanais. 2004. “Elevated Depressive Symptoms among Caregiving Grandparents.” *Health Services Research* 39(6):1671–90.

Bryson, Ken and Lynne M. Casper. 1999. “Coresident Grandparents and Grandchildren.” *Current Population Reports*: P23-198. U.S. Census Bureau.

Burn, Katherine and Cassandra Szoeke. 2015. “Grandparenting Predicts Late-Life Cognition: Results from the Women's Healthy Ageing Project.” *Maturitas* 81(2):317–22.

Burnette, Denise. 1999. “Custodial Grandparents in Latino Families: Patterns of Service Use and Predictors of Unmet Needs.” *Social Work* 44(1):22–34.

Burnette, Denise, Juanjuan Sun, and Fei Sun. 2013. “A Comparative Review of Grandparent Care of Children in the U.S. And China.” *Ageing International* 38(1):43–57.

Casper, Lynne M. and Suzanne M. Bianchi. 2002. *Continuity and Change in the American Family*. Thousand Oaks, CA: SAGE.

Casper, Lynne M., Sandra M. Florian, C. Brady Potts, and Peter D. Brandon. 2016. "Portrait of American Grandparent Families." Pp. 109–32 in *Grandparenting in the United States*, edited by Madonna Harrington Meyer and Ynesse Abdul-Malak. Amityville, New York: Baywood PUblishing Company.

Chen, Feinian and Guangya Liu. 2012. "The Health Implications of Grandparents Caring for Grandchildren in China." *The Journals of Gerontology: Series B* 67B(1):99–112.

Chen, Feinian, Christine A. Mair, Luoman Bao, and Yang Claire Yang. 2015. "Race/Ethnic Differentials in the Health Consequences of Caring for Grandchildren for Grandparents." *The Journals of Gerontology: Series B* 70(5):793–803.

Choi, Seung-won Emily. 2020. "Grandparenting and Mortality: How Does Race-Ethnicity Matter?" *Journal of Health and Social Behavior* 61(1):96–112.

Coall, David A. and Ralph Hertwig. 2011. "Grandparental Investment: A Relic of the Past or a Resource for the Future?" *Current Directions in Psychological Science* 20(2):93–8.

Cohen, Philip N. and Lynne M. Casper. 2002. "In Whose Home? Multigenerational Families in the United States, 1998–2000." *Sociological Perspectives* 45(1):1–20.

Deaton, Angus and Arthur A. Stone. 2014. "Grandpa and the Snapper: The Well-Being of the Elderly Who Live with Children." Pp. 283–300 in *Discoveries in the Economics of Aging*, edited by David A. Wise. Chicago, IL: University of Chicago Press.

Di Gessa, Giorgio, Karen Glaser, and Anthea Tinker. 2016a. "The Health Impact of Intensive and Nonintensive Grandchild Care in Europe: New Evidence from Share." *The Journals of Gerontology: Series B* 71(5):867–79.

Di Gessa, Giorgio, Karen Glaser, and Anthea Tinker. 2016b. "The Impact of Caring for Grandchildren on the Health of Grandparents in Europe: A Lifecourse Approach." *Social Science & Medicine* 152:166–75.

Dunifon, Rachel E., Kathleen M. Ziol-Guest, and Kimberly Kopko. 2014. "Grandparent Coresidence and Family Well-Being: Implications for Research and Policy." *The Annals of the American Academy of Political and Social Science* 654(1):110–26.

Dunifon, Rachel E., Kelly A. Musick, and Christopher E. Near. 2020. "Time with Grandchildren: Subjective Well-Being among Grandparents Living with Their Grandchildren." *Social Indicators Research* 148(2):681–702.

Ellis, Renee R. and Tavia Simmons. 2014. "Coresident Grandparents and Their Grandchildren: 2012." *Current Population Reports*: P20-576. U.S. Census Bureau.

Ellwardt, Lea, Karsten Hank, and Carlos F. Mendes de Leon. 2021. "Grandparenthood and Risk of Mortality: Findings from the Health and Retirement Study." *Social Science & Medicine* 268:113371.

Emick, Michelle, A. and Bert Hayslip, Jr. 1999. "Custodial Grandparenting: Stresses, Coping Skills, and Relationships with Grandchildren." *The International Journal of Aging and Human Development* 48(1):35–61.

Finlay, Keith and Katie R. Genadek. 2021. "Measuring All-Cause Mortality with the Census Numident File." *American Journal of Public Health* 111(S2):S141–48.

Florian, Sandra M. and Lynne M. Casper. 2015. "Coresident Grandparent Caring for Grandchildren: Variations by Race/Ethnicity, Gender, and Family Structure." Paper presented at the 2016 Annual Meeting of the Population Association of America, Washington, D.C.

Fuller-Thomson, Esme, Meredith Minkler, and Diane Driver. 1997. “A Profile of Grandparents Raising Grandchildren in the United States.” *The Gerontologist* 37(3):406–11.

Fuller-Thomson, Esme and Meredith Minkler. 2007. “Central American Grandparents Raising Grandchildren.” *Hispanic Journal of Behavioral Sciences* 29(1):5–18.

Glick, Jennifer E. and Jennifer Van Hook. 2002. “Parents' Coresidence with Adult Children: Can Immigration Explain Racial and Ethnic Variation?" *Journal of Marriage and Family* 64(1):240–53.

Goode, William J. 1960. “A Theory of Role Strain.” *American Sociological Review* 25(1):483–96.

Goodman, Catherine Chase and Merril Silverstein. 2002. “Grandmothers Raising Grandchildren: Family Structure and Well-Being in Culturally Diverse Families.” *The Gerontologist* 42(5):676–89.

Goodman, Catherine Chase and Merril Silverstein. 2006. “Grandmothers Raising Grandchildren: Ethnic and Racial Differences in Well-Being among Custodial and Coparenting Families.” *Journal of Family Issues* 27(11):1605–26.

Hayslip, Bert, Jr. and Patricia L. Kaminski. 2005. “Grandparents Raising Their Grandchildren: A Review of the Literature and Suggestions for Practice.” *The Gerontologist* 45(2):262–69.

Hayslip, Bert, Jr., Christine A. Fruhauf, and Megan L. Dolbin-MacNab. 2019. “Grandparents Raising Grandchildren: What Have We Learned over the Past Decade?" *The Gerontologist* 59(3):e152–63.

Hilbrand, Sonja, David A. Coall, Denis Gerstorf, and Ralph Hertwig. 2017. “Caregiving within and Beyond the Family Is Associated with Lower Mortality for the Caregiver: A Prospective Study.” *Evolution and Human Behavior* 38(3):397–403.

Hughes, Mary Elizabeth, Linda J. Waite, Tracey A. LaPierre, and Ye Luo. 2007. “All in the Family: The Impact of Caring for Grandchildren on Grandparents' Health.” *The Journals of Gerontology: Series B* 62(2):S108–19.

Jendrek, Margaret Platt. 1993. “Grandparents Who Parent Their Grandchildren: Effects on Lifestyle.” *Journal of Marriage and Family* 55(3):609–21.

Jimenez, Jillian. 2002a. “The History of Grandmothers in the African‐American Community.” *Social Service Review* 76(4):523–51.

Jimenez, Jillian. 2002b. “The History of Grandmothers in the African-American Community.” *Social Service Review* 76(4):523–51.

Kamo, Yoshinori. 2000. “Racial and Ethnic Differences in Extended Family Households.” *Sociological Perspectives* 43(2):211–29.

Kataoka-Yahiro, Merle R., Clementina Ceria, and Marian Yoder. 2004. “Grandparent Caregiving Role in Filipino American Families.” *Journal of Cultural Diversity* 11(3):110–17.

Kataoka-Yahiro, Merle R. 2010. “Filipino American Grandparent Caregivers' Roles, Acculturation, and Perceived Health Status.” *Journal of Cultural Diversity* 17(1):24–33.

Kelley, Susan J. 1993. “Caregiver Stress in Grandparents Raising Grandchildren.” *Image: the Journal of Nursing Scholarship* 25(4):331–37.

Kim, Jeung Hyun. 2022. “Grandparenting, Filial Piety, and Well-Being of Chinese-American Older Adults.” *Current Sociology* 70(6):860–79.

Livingston, Gretchen. 2013. “At Grandmother’s House We Stay: One-in-Ten Children Are Living with a Grandparent.” Washington, DC: Pew Research Center. Retrieved September 25, 2025 (https://www.pewresearch.org/social-trends/2013/09/04/at-grandmothers-house-we-stay).

Luo, Ye, Tracey A. LaPierre, Mary Elizabeth Hughes, and Linda J. Waite. 2012. "Grandparents Providing Care to Grandchildren: A Population-Based Study of Continuity and Change." *Journal of Family Issues* 33(9):1143–67.

Mendoza, Angela N., Christine A. Fruhauf, and David MacPhee. 2020. "Grandparent Caregivers' Resilience: Stress, Support, and Coping Predict Life Satisfaction." *The International Journal of Aging and Human Development* 91(1):3–20.

Minkler, Meredith, Kathleen M. Roe, and Marilyn Price. 1992. "The Physical and Emotional Health of Grandmothers Raising Grandchildren in the Crack Cocaine Epidemic." *The Gerontologist* 32(6):752–61.

Minkler, Meredith, Esme Fuller-Thomson, Doriane Miller, and Diane Driver. 1997. "Depression in Grandparents Raising Grandchildren: Results of a National Longitudinal Study." *Archives of Family Medicine* 6(5):445–52.

Minkler, Meredith. 1999. "Intergenerational Households Headed by Grandparents: Contexts, Realities, and Implications for Policy." *Journal of Aging Studies* 13(2):199–218.

Minkler, Meredith and Esme Fuller-Thomson. 2000. "Second Time around Parenting: Factors Predictive of Grandparents Becoming Caregivers for Their Grandchildren." *The International Journal of Aging and Human Development* 50(3):185–200.

Minkler, Meredith and Esme Fuller-Thomson. 2001. "Physical and Mental Health Status of American Grandparents Providing Extensive Child Care to Their Grandchildren." *Journal of the American Medical Women's Association* 56(4):199–205.

Minkler, Meredith and Esme Fuller-Thomson. 2005. "African American Grandparents Raising Grandchildren: A National Study Using the Census 2000 American Community Survey." *The Journals of Gerontology: Series B* 60(2):S82–S92.

Moen, Phyllis, Julie Robison, and Donna Dempster-McClain. 1995. “Caregiving and Women's Well-Being: A Life Course Approach.” *Journal of Health and Social Behavior* 36(3):259–73.

Pan, Xi, Ye Luo, and Nicholas Joseph Bishop. 2022. “Cognitive Function among Noncustodial Grandparents in China and the United States: A Cross-National Perspective.” *The International Journal of Aging and Human Development* 95(1):18–41.

Pearlin, Leonard I., Joseph T. Mullan, Shirley J. Semple, and Marilyn M. Skaff. 1990. “Caregiving and the Stress Process: An Overview of Concepts and Their Measures.” *The Gerontologist* 30(5):583–94.

Pruchno, Rachel. 1999. “Raising Grandchildren: The Experiences of Black and White Grandmothers.” *The Gerontologist* 39(2):209–21.

Pruchno, Rachel A. and Dorothy McKenney. 2002. “Psychological Well-Being of Black and White Grandmothers Raising Grandchildren: Examination of a Two-Factor Model.” *The Journals of Gerontology: Series B* 57(5):444–52.

Simmons, Tavia and Jane Lawler Dye. 2001. “Grandparents Living with Grandchildren: 2000.” *Census 2000 Brief*: C2KBR-31. U.S. Census Bureau.

Sneed, Rodlescia S. and Richard Schulz. 2019. “Grandparent Caregiving, Race, and Cognitive Functioning in a Population-Based Sample of Older Adults.” *Journal of Aging and Health* 31(3):415–38.

Szinovacz, Maximiliane E., Stanley DeViney, and Maxine P. Atkinson. 1999. “Effects of Surrogate Parenting on Grandparents' Well-Being.” *The Journals of Gerontology: Series B* 54B(6):S376–88.

Tang, Fengyan, Heejung Jang, and Elizabeth Mulvaney. 2019. “Asian American Grandparents Caring for Grandchildren: Findings from the Census 2010-2012 American Community Survey.” *GrandFamilies: The Contemporary Journal of Research, Practice and Policy* 5(2):13–24.

Waldrop, Deborah and Joseph Weber. 2001. “From Grandparent to Caregiver: The Stress and Satisfaction of Raising Grandchildren.” *Families in Society: The Journal of Contemporary Social Services* 82(5):461–72.

Whitley, Deborah M. and Esme Fuller-Thomson. 2017. “African–American Solo Grandparents Raising Grandchildren: A Representative Profile of Their Health Status.” *Journal of Community Health* 42(2):312–23.

Xu, Hongwei. 2022. “Grandparenting and Cognitive Functioning in China.” *Population Research and Policy Review* 41(1):285–316.

Xu, Hongwei, Jinyu Liu, Zhenmei Zhang, and Lydia Li. 2022. “Sandwiched Grandparents and Biological Health Risks in China.” *Journal of Health and Social Behavior* 63(3):410–27.

Table 1. Coresident Grandparenting Status among U.S. Adults Aged 50 Years or Older.

	White	Black	Hispanic	Asian
All adults aged 50 or older				
Not coresident grandparents (%)	96.8	86.2	83.4	86.0
Coresident grandparents (%)	3.2	13.8	16.6	14.1
Weighted *n*	53,000,000	5,473,000	3,759,000	1,795,000
Coresident grandparents				
Coresident only (%)	60.9	50.8	68.4	81.8
Primary caregivers (%)	39.1	49.2	31.6	18.2
Weighted *n*	1,702,000	754,000	624,000	252,000
Primary caregivers				
Caregiving < 1 year (%)	18.6	13.1	18.6	21.5
Caregiving 1–2 years (%)	19.8	15.6	19.8	23.3
Caregiving 3–4 years (%)	15.5	13.3	15.5	18.3
Caregiving 5+ years (%)	46.1	57.9	46.1	36.9
Weighted *n*	665,000	371,000	197,000	46,000

Note. Weighted sample sizes are rounded to comply with disclosure requirements from the U.S. Census Bureau.

Table 2. Characteristics of Non-Hispanic White U.S. Adults Aged 50 or Older in 2000 by Coresident Grandparenting Status.

Death Rate during 2000 – 2019	Noncoresident	Coresident				
		Coresident only	Primary caregiver			
			< 1 year	1–2 years	3–4 years	5+ years
% Whites died	56.0	60.0	45.4	44.7	46.9	53.6
% Blacks died	57.5	56.0	46.1	45.8	48.5	53.7
% Hispanics died	43.6	41.9	32.0	30.9	34.7	41.3
% Asians died	33.0	45.8	30.7	30.2	30.4	40.8

Table 3. Weighted Cox Proportional Hazard Models of Associations between Coresident Grandparenting and Mortality during 2000–2019: U.S. Adults Aged 50 or Older in 2000.

	White		Black		Hispanic		Asian	
Independent variables in 2000	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Coresident grandparenting (ref: noncoresident)								
Coresident only	1.213***	1.100***	1.018	1.014	0.888***	1.005	0.875***	0.945***
Primary caregiver < 1 year	1.226***	1.149***	0.960*	0.945**	0.933	0.983	0.920	0.921†
Primary caregiver 1–2 years	1.213***	1.148***	0.964†	0.945**	0.938*	0.999	0.871***	0.895**
Primary caregiver 3–4 years	1.232***	1.172***	1.009	1.001	1.001	1.048*	0.867*	0.874*
Primary caregiver 5+ years	1.298***	1.204***	1.047***	1.005	1.039*	1.053*	0.931†	0.928**
Basic demographics								
Age (mean-centered)	1.114***	1.097***	1.089***	1.073***	1.110***	1.094***	1.119***	1.103***
Age (mean-centered) squared	0.999***	0.999***	0.999***	0.999***	0.999***	0.999***	0.999***	0.999***
Male (ref: female)	1.394***	1.567***	1.425***	1.566***	1.406***	1.561***	1.393***	1.508***
Household structure								
Married (ref: not)		0.837***		0.884***		0.882***		0.884***
Household head (ref: not)		0.900***		0.890***		1.020†		1.167***
Children in household		0.954***		0.985***		0.963***		1.005
Adults in household		1.036***		1.017***		0.991**		1.006**

Socioeconomic status				
Education (ref: < high school)				
High school	0.927***	0.937***	0.958***	0.970**
Some college	0.873***	0.886***	0.900***	0.914***
College or above	0.742***	0.821***	0.858***	0.829***
Employed (ref: unemployed)	0.749***	0.691***	0.721***	0.756***
Income-to-poverty ratio	0.950***	0.948***	0.954***	0.969***
Homeowner (ref: not)	0.837***	0.871***	0.909***	0.934***
Acculturation				
Citizenship (ref: born citizen)				
Naturalized citizen	0.853***	0.663***	0.800***	0.916***
Not citizen	0.735***	0.524***	0.685***	0.760***
Speak English at home (ref: not)	1.116***	1.054**	1.042***	1.053*
Health status				
Any disability (ref: none)	1.584***	1.392***	1.444***	1.382***
Weighted n	53,000,000	5,473,000	3,759,000	1,795,000

Note. ref = reference. Significance based on robust standard errors clustered at the state level. All models adjusted for state-fixed effects. Weighted sample sizes rounded to comply with disclosure requirements from the U.S. Census Bureau. Based on restricted data from the weighted 2000 U.S. census long-form sample linked to the 2019 Numident file.

†$p < .100$. *$p < .05$. **$p < .01$. ***$p < .001$ in two-tailed tests.

Table 4. Weighted Cox Proportional Hazard Models of Associations between Coresident Grandparenting and Mortality during 2000–2019: U.S. Coresident Grandparents Aged 50 or Older in 2000.

	White		Black		Hispanic		Asian	
Independent variables in 2000	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Coresident grandparenting								
(ref: coresident only)								
Primary caregiver < 1 year	0.936***	0.968*	0.929***	0.917***	0.995	0.930	0.995	0.947
Primary caregiver 1–2 years	0.923***	0.964**	0.931**	0.916***	0.997	0.939*	0.942	0.925†
Primary caregiver 3–4 years	0.944***	0.988	0.976	0.969†	1.073†	0.991	0.938	0.898†
Primary caregiver 5+ years	1.003	1.017	1.016	0.971**	1.126***	0.987	1.038	0.961*
Basic demographics								
Age (mean-centered)	1.097***	1.083***	1.084***	1.072***	1.099***	1.089***	1.105***	1.099***
Age (mean-centered) squared	0.999***	0.999***	0.999***	0.999***	0.999***	0.999***	0.999***	0.999***
Male (ref: female)	1.297***	1.450***	1.331***	1.487***	1.325***	1.448***	1.391***	1.423***
Household structure								
Married (ref: not)		0.865***		0.950***		0.917***		0.939**
Household head (ref: not)		0.907***		0.904***		1.076***		1.143**
Children in household		0.999		1.010*		1.003		1.031***
Adults in household		1.030***		1.023***		0.994		1.033***
Socioeconomic status								

Education (ref: < high school)				
High school	0.913***	0.916***	0.933***	0.937**
Some college	0.833***	0.896***	0.881***	0.950†
College or above	0.753***	0.871***	0.874**	0.894***
Employed (ref: unemployed)	0.737***	0.687***	0.727***	0.844***
Income-to-poverty ratio	0.937***	0.946***	0.968***	0.961***
Homeowner (ref: not)	0.851***	0.871***	0.949***	0.970†
Acculturation				
Citizenship (ref: born citizen)				
Naturalized citizen	0.766***	0.664***	0.740***	0.944
Not citizen	0.505***	0.458***	0.600***	0.710***
Speak English at home (ref: not)	1.178***	1.111***	1.009	1.114*
Health status				
Any disability (ref: none)	1.648***	1.434***	1.499***	1.422***
Weighted n	1,702,000	754,000	624,000	252,000

Note. ref = reference. Significance based on robust standard errors clustered at the state level. All models adjusted for state-fixed effects. Weighted sample sizes rounded to comply with disclosure requirements from the U.S. Census Bureau. Based on restricted data from the weighted 2000 U.S. census long-form sample linked to the 2019 Numident file.

†$p < .100$. *$p < .05$. **$p < .01$. ***$p < .001$ in two-tailed tests.

Appendix Table A1. Characteristics of Non-Hispanic White U.S. Adults Aged 50 or Older in 2000 by Coresident Grandparenting Status

	Noncoresident	Coresident				
		Coresident only	Primary caregiver			
			< 1 year	1–2 years	3–4 years	5+ years
Basic demographics						
Age (mean)	64.9	64.4	58.8	58.7	59.2	60.7
Female (%)	54.1	63.7	53.3	53.3	54.5	57.1
Household structure						
Currently married (%)	66.7	50.8	76.3	76.0	77.0	76.3
Household head or spouse (%)	94.8	64.1	96.1	96.2	96.8	96.7
Children in household (*mean*)	0.1	1.7	1.5	1.5	1.5	1.5
Adults in household (mean)	2.0	3.3	2.7	2.7	2.7	2.5
Socioeconomic status						
Education (%)						
< High school	19.9	28.8	24.5	23.8	24.8	31.8
High school	32.3	38.3	34.6	35.4	35.3	35.3
Some college	24.4	21.5	27.2	27.5	27.2	23.7
College or above	23.4	11.4	13.7	13.3	12.7	9.3
Currently employed (%)	42.2	37.9	55.6	56.6	54.2	47.3
Income-to-poverty ratio (mean)	3.6	3.4	3.0	3.0	3.0	2.8
Homeowner (%)	85.4	86.7	84.7	86.0	86.2	85.4
Acculturation						
Citizenship status (%)						
Born citizen	94.9	91.6	96.3	96.2	96.1	96.7
Naturalized citizen	3.9	4.9	2.3	2.4	2.6	2.3
Foreign national	1.2	3.5	1.4	1.4	1.3	1.1
Speak English at home (%)	93.9	90.2	94.5	94.8	94.6	95.0
Health						
Any disability (%)	30.7	39.8	33.7	32.7	33.3	37.5
Weighted n	51,300,000	1,037,000	124,000	132,000	103,000	307,000

Note. Weighted sample sizes are rounded to comply with disclosure requirements from the U.S. Census Bureau.

Appendix Table A2. Characteristics of Non-Hispanic Black U.S. Adults Aged 50 or Older in 2000 by Coresident Grandparenting Status

	Noncoresident	Coresident				
		Coresident only	Primary caregiver			
			< 1 year	1–2 years	3–4 years	5+ years
Basic demographics						
Age (*M*)	63.5	62.9	59.0	59.1	59.6	61.1
Female (%)	57.1	70.2	64.2	65.3	67.9	70.6
Household structure						
Currently married (%)	44.9	41.0	53.6	53.5	51.8	50.4
Household head or spouse (%)	88.8	75.4	95.1	95.1	95.7	96.2
Children in household (mean)	0.2	1.8	1.8	1.8	1.9	1.9
Adults in household (mean)	2.1	3.2	2.7	2.7	2.7	2.5
Socioeconomic status						
Education (%)						
< High school	39.7	44.8	39.5	40.3	42.3	47.8
High school	26.1	29.5	27.9	29.1	28.8	27.4
Some college	20.7	18.4	23.1	22.2	20.8	18.7
College or above	13.4	7.3	9.5	8.4	8.1	6.1
Currently employed (%)	39.1	35.0	47.6	47.3	45.4	38.9
Income-to-poverty ratio (mean)	2.8	2.6	2.3	2.3	2.3	2.1
Homeowner (%)	67.7	72.4	68.0	68.1	69.1	68.4
Acculturation						
Citizenship status (%)						
Born citizen	93.9	89.9	93.6	93.7	94.2	96.3
Naturalized citizen	4.2	5.7	4.5	4.1	4.0	2.7
Foreign national	1.9	4.5	2.0	2.2	1.8	1.0
Speak English at home (%)	96.2	94.6	95.4	95.9	96.6	97.5
Health						
Any disability (%)	44.5	46.1	43.9	43.6	43.9	47.4
Weighted n	4,719,000	383,000	48,500	58,000	49,500	215,000

Note. Weighted sample sizes are rounded to comply with disclosure requirements from the U.S. Census Bureau.

Appendix Table A3. Characteristics of Hispanic U.S. Adults Aged 50 or Older in 2000 by Coresident Grandparenting Status

	Noncoresident	Coresident				
		Coresident only	Primary caregiver			
			< 1 year	1–2 years	3–4 years	5+ years
Basic demographics						
Age (*M*)	62.1	62.6	58.2	57.8	58.4	60.6
Female (%)	54.0	61.6	55.3	55.1	56.0	61.1
Household structure						
Currently married (%)	62.5	57.1	70.3	72.1	71.5	68.5
Household head or spouse (%)	86.7	55.2	86.9	88.4	89.5	89.6
Children in household (mean)	0.3	2.1	2.0	2.0	2.0	2.0
Adults in household (mean)	2.5	3.9	3.5	3.5	3.4	3.2
Socioeconomic status						
Education (%)						
< High school	53.6	72.6	65.1	63.1	64.4	69.1
High school	19.9	15.2	17.0	18.7	18.6	16.9
Some college	16.3	8.6	12.9	13.3	12.6	10.3
College or above	10.2	3.6	5.0	5.0	4.5	3.7
Currently employed (%)	40.7	30.9	44.3	46.5	44.0	35.9
Income-to-poverty ratio (mean)	2.8	2.4	2.2	2.2	2.2	2.1
Homeowner (%)	67.0	69.3	65.0	66.2	67.0	66.5
Acculturation						
Citizenship status (%)						
Born citizen	51.1	33.6	43.9	46.6	48.2	56.0
Naturalized citizen	30.0	27.3	24.4	24.5	23.4	21.9
Foreign national	18.9	39.1	31.7	28.9	28.5	22.1
Speak English at home (%)	15.6	10.2	11.0	11.9	11.9	11.7
Health						
Any disability (%)	39.3	40.9	40.8	37.0	41.2	43.7
Weighted n	3,135,000	427,000	36,500	39,000	30,500	91,000

Note. Weighted sample sizes are rounded to comply with disclosure requirements from the U.S. Census Bureau.

Appendix Table A4. Characteristics of Asian U.S. Adults Aged 50 or Older in 2000 by Coresident Grandparenting Status

	Noncoresident	Coresident				
		Coresident only	Primary caregiver			
			< 1 year	1–2 years	3–4 years	5+ years
Basic demographics						
Age (*M*)	61.1	67.2	60.9	61.1	61.5	64.7
Female (%)	53.5	61.1	59.3	59.7	60.0	62.8
Household structure						
Currently married (%)	76.2	63.4	78.5	75.6	76.9	70.3
Household head or spouse (%)	86.5	29.0	64.0	65.8	62.9	63.4
Children in household (mean)	0.3	1.9	1.8	1.7	1.8	2.0
Adults in household (mean)	2.7	4.1	4.1	3.9	3.9	3.7
Socioeconomic status						
Education (%)						
< High school	25.8	50.9	42.3	43.4	45.3	48.6
High school	18.7	19.5	22.0	20.1	20.8	19.4
Some college	18.1	11.7	13.0	14.8	15.8	15.0
College or above	37.3	17.9	22.7	21.7	18.2	17.0
Currently employed (%)	51.1	21.5	37.5	39.2	39.2	32.5
Income-to-poverty ratio (mean)	3.6	3.3	3.0	3.1	3.2	2.9
Homeowner (%)	72.8	79.1	68.9	70.7	74.4	76.3
Acculturation						
Citizenship status (%)						
Born citizen	16.9	5.8	5.3	6.1	8.7	11.0
Naturalized citizen	58.7	45.1	51.4	50.3	49.0	52.3
Foreign national	24.4	49.1	43.4	43.6	42.3	36.8
Speak English at home (%)	18.2	7.7	9.0	9.9	13.1	13.5
Health						
Any disability (%)	27.9	39.4	34.2	31.0	34.2	38.0
Weighted n	1,543,000	206,000	9,800	10,500	8,400	17,000

Note. Weighted sample sizes are rounded to comply with disclosure requirements from the U.S. Census Bureau.

Appendix Table A5. Weighted Gompertz Models of Associations between Coresident Grandparenting and Mortality during 2000–2019: U.S. Adults Aged 50 or Older in 2000.

Independent variables in 2000	White	Black	Hispanic	Asian
Coresident grandparenting (ref: noncoresident)				
Coresident only	1.101***	1.015	1.005	0.945***
Primary caregiver < 1 year	1.149***	0.944**	0.983	0.921†
Primary caregiver 1–2 years	1.148***	0.945**	0.999	0.895**
Primary caregiver 3–4 years	1.172***	1.001	1.048*	0.874*
Primary caregiver 5+ years	1.205***	1.005	1.053*	0.928**
Basic demographic	Yes	Yes	Yes	Yes
Household structure	Yes	Yes	Yes	Yes
Socioeconomic status	Yes	Yes	Yes	Yes
Acculturation	Yes	Yes	Yes	Yes
Health status	Yes	Yes	Yes	Yes
Ancillary parameter gamma	0.0003***	0.0002***	0.0003***	0.0003***
Weighted n	53,000,000	5,476,000	3,759,000	1,795,000

Note. ref = reference. Significance based on robust standard errors clustered at the state level. All models adjusted for state-fixed effects. Weighted sample sizes rounded to comply with disclosure requirements from the U.S. Census Bureau. Based on restricted data from the weighted 2000 U.S. census long-form sample linked to the 2019 Numident file.
†$p < .100$. *$p < .05$. **$p < .01$. ***$p < .001$ in two-tailed tests.

Appendix Table A6. Weighted Gompertz Models of Associations between Coresident Grandparenting and Mortality during 2000–2019: U.S. Coresident and Custodial Grandparents Aged 50 or Older in 2000.

Independent variables in 2000	White	Black	Hispanic	Asian
Coresident grandparenting (ref: coresident only)				
Primary caregiver < 1 year	0.968*	0.917***	0.930	0.947
Primary caregiver 1–2 years	0.964**	0.916***	0.938*	0.925†
Primary caregiver 3–4 years	0.988	0.969†	0.990	0.898†
Primary caregiver 5+ years	1.017	0.971**	0.987	0.961*
Basic demographic	Yes	Yes	Yes	Yes
Household structure	Yes	Yes	Yes	Yes
Socioeconomic status	Yes	Yes	Yes	Yes
Acculturation	Yes	Yes	Yes	Yes
Health status	Yes	Yes	Yes	Yes
Ancillary parameter gamma	0.0003***	0.0002***	0.0002***	0.0002***
Weighted n	1,702,000	754,000	624,000	252,000

Note. ref = reference. Significance based on robust standard errors clustered at the state level. All models adjusted for state-fixed effects. Weighted sample sizes rounded to comply with disclosure requirements from the U.S. Census Bureau. Based on restricted data from the weighted 2000 U.S. census long-form sample linked to the 2019 Numident file.
†$p < .100$. *$p < .05$. **$p < .01$. ***$p < .001$ in two-tailed tests.

Appendix Table A7. Weighted Cox Models of Associations between Coresident Grandparenting and Mortality during 2000–2019, Stratified by Nativity: U.S. Adults Aged 50 or Older in 2000.

	White		Black		Hispanic		Asian	
Independent variables in 2000	Native	Foreign	Native	Foreign	Native	Foreign	Native	Foreign
Coresident grandparenting (ref: noncoresident)								
Coresident only	1.117***	0.954†	1.018	0.967	1.047**	0.961	1.018	0.926***
Primary caregiver < 1 year	1.148***	1.033	0.937**	1.107	1.010	0.939	1.314†	0.884*
Primary caregiver 1–2 years	1.149***	1.001	0.942**	0.960	1.010	0.961	1.003	0.867**
Primary caregiver 3–4 years	1.170***	1.093	0.999	0.983	1.027	1.042	1.098	0.836**
Primary caregiver 5+ years	1.201***	1.103*	1.000	0.999	1.061*	0.988	0.903*	0.923*
Basic demographic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household structure	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Socioeconomic status	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Acculturation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Health status	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weighted n	50,280,000	2,717,000	5,128,000	345,000	1,845,000	1,914,000	277,000	1,518,000

Note. ref = reference. Significance based on robust standard errors clustered at the state level. All models adjusted for state-fixed effects. Weighted sample sizes rounded to comply with disclosure requirements from the U.S. Census Bureau. Based on restricted data from the weighted 2000 U.S. census long-form sample linked to the 2019 Numident file.
†$p < .100$. *$p < .05$. **$p < .01$. ***$p < .001$ in two-tailed tests.

Appendix Table A8. Weighted Cox Models of Associations between Coresident Grandparenting and Mortality during 2000–2019, Stratified by Nativity: U.S. Coresident and Custodial Grandparents Aged 50 or Older in 2000.

	White		Black		Hispanic		Asian	
Independent variables in 2000	Native	Foreign	Native	Foreign	Native	Foreign	Native	Foreign
Coresident grandparenting (ref: coresident only)								
Primary caregiver < 1 year	0.967*	0.992	0.910***	1.117†	0.940†	0.923	1.262	0.928†
Primary caregiver 1–2 years	0.965**	0.959	0.914***	0.966	0.940	0.943	0.987	0.914†
Primary caregiver 3–4 years	0.986	1.032	0.969†	0.985	0.963	1.026	1.076	0.879*
Primary caregiver 5+ years	1.015	1.050	0.970**	0.982	0.989	0.982	0.816**	0.986
Basic demographic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household structure	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Socioeconomic status	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Acculturation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Health status	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weighted n	1,591,000	111,000	697,000	56,000	243,000	380,000	16,000	236,000

Note. ref = reference. Significance based on robust standard errors clustered at the state level. All models adjusted for state-fixed effects. Weighted sample sizes rounded to comply with disclosure requirements from the U.S. Census Bureau. Based on restricted data from the weighted 2000 U.S. census long-form sample linked to the 2019 Numident file.
†$p < .100$. *$p < .05$. **$p < .01$. ***$p < .001$ in two-tailed tests.